

Experiment 4.2

Determination and plotting of Electric Field Intensity due to Line Charge

Code File Name: Exp4_2_LineCharge_EField.m

Code

```
clc;
clear;
close all;

L = 10;
Q = 50e-6;

lambda = Q/L;

disp('Line charge density:');
disp(lambda);

epsilon = 8.854e-12;

r = linspace(0.1,5,100);

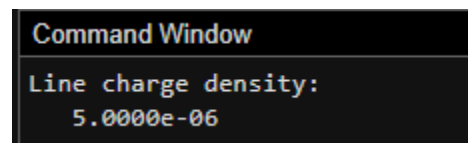
E = lambda./(2*pi*epsilon*r);

figure;
plot(r,E,'LineWidth',2);
grid on;

xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Line Charge');
```

Command Prompt

Line charge density:
5.0000e-06



The screenshot shows a MATLAB Command Window with a black background and white text. The title bar reads "Command Window". The output displayed is "Line charge density:" followed by "5.0000e-06" on the next line.

Output Graph

